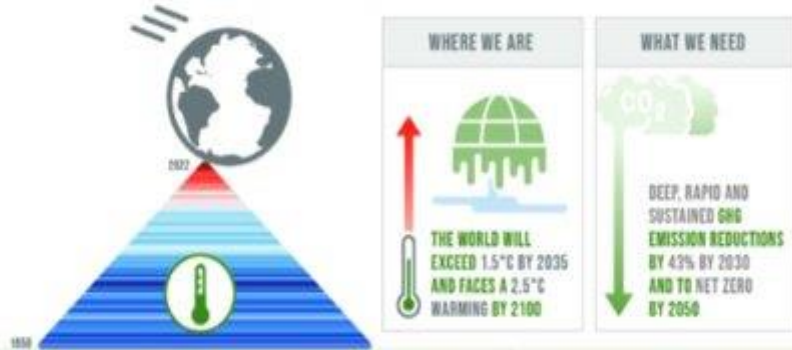


Part 1: Climate Action

NAME OF STUDENTS

EARTH'S TIPPING POINT

STANDING AT THE BRINK OF CLIMATE CALAMITY



BILLIONS TO TRILLIONS:

GLOBAL CLIMATE FINANCE FLOWS REACHED AN ANNUAL AVERAGE OF \$803 BILLION IN 2019-2020

HOWEVER, DEVELOPING COUNTRIES REQUIRE NEARLY \$8 TRILLION BY 2030



THE RATE OF SEA-LEVEL RISE HAS DOUBLED IN THE LAST DECADE



HIGHLY VULNERABLE REGIONS

EXPERIENCE 15X HIGHER MORTALITY RATES FROM DISASTERS COMPARED TO VERY LOW VULNERABILITY REGIONS (2010-2020)



INTRODUCTION

Climate change is a problem that affects everyone worldwide. It is imperative that there be a climate action plan created to solve the issues that have already caused so much damage. In addition, a plan to protect our future livelihood.



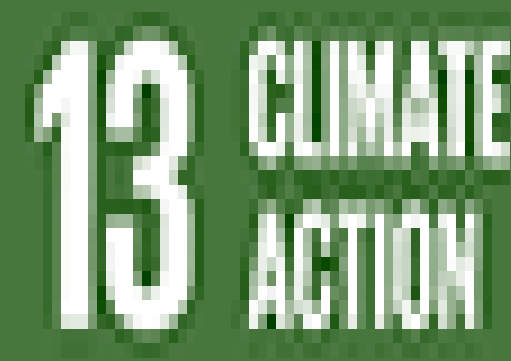
PROBLEM DESCRIPTION: SDG 13

As a result of the rising temperatures and extreme weather conditions due to climate change, copious amounts of Canadians have been affected by natural disasters that have destroyed their livelihoods.

The objective of Sustainable Development Goal (SDG) 13 is to "Take urgent action to combat climate change and its impacts." (United Nations, -c).

PROBLEM DESCRIPTION: SDG 13 Targets

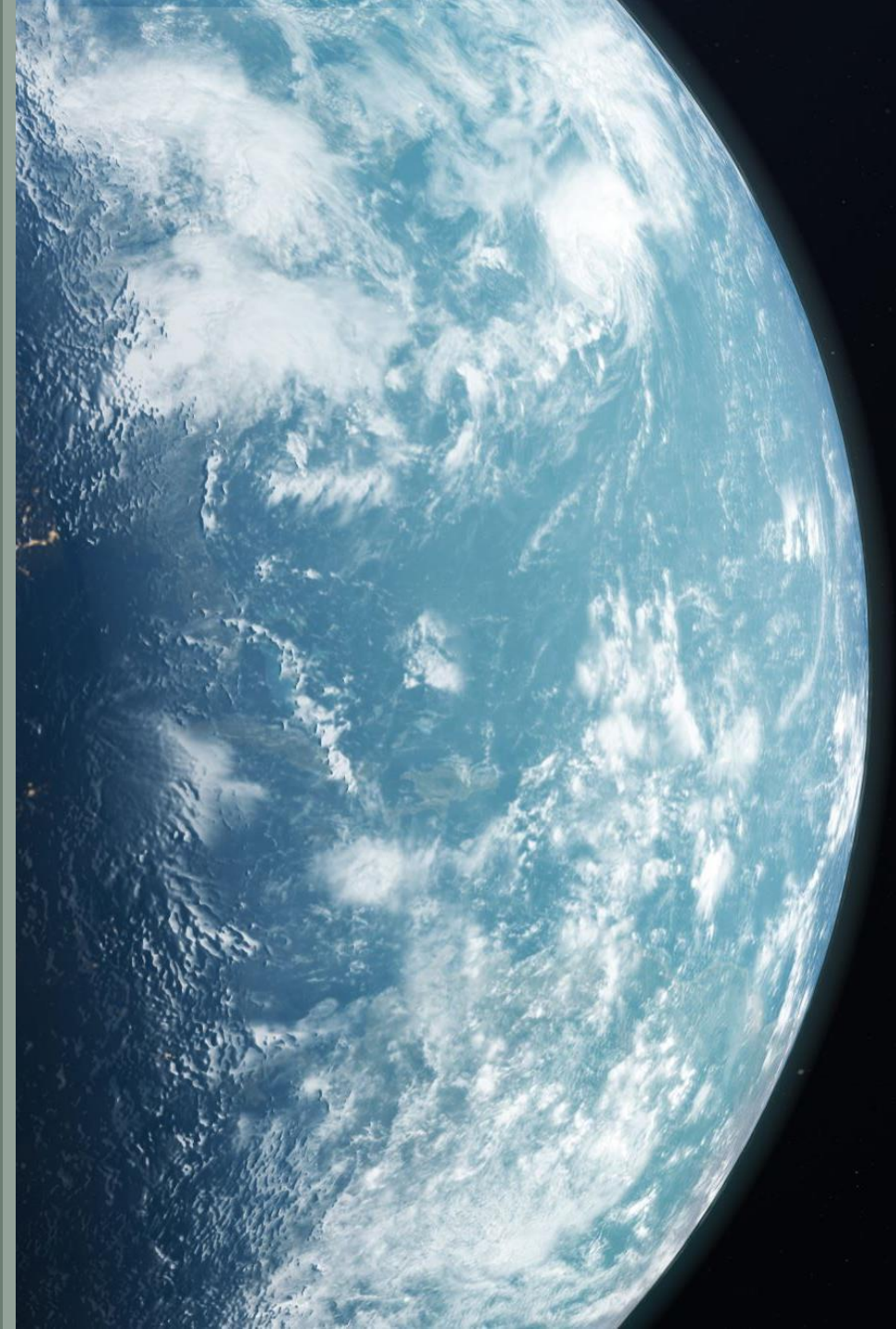
Targets include enhancing ability to adapt and be resilient to risks associated with climate change and natural disasters (United Nations, -b). Including strategy, planning, and national policies that address climate change (United Nations, -b). Increasing human and institutional capacity, knowledge, and education on early warning, impact reduction, adaptation, and mitigation of climate change (United Nations, -b). By 2020, \$100 billion a year should be raised to help developing nations combat climate change through mitigation measures and open implementation (United Nations, -b). With an emphasis on vulnerable populations like women, youth, and marginalised groups, it seeks to quickly fully operationalize the Green Climate Fund and strengthen the capacity of small island developing states and least developed countries (United Nations, -b).



PROBLEM DESCRIPTION:

World Situation

Climate change remains a serious hazard on a global scale. The number of people affected by disasters per 100,000 people has increased by more than two-thirds, from 1,169 in 2005–2014 to 1,980 in 2013–2022 (United Nations, -a). According to the Intergovernmental Panel on Climate Change (IPCC), since the late 19th century, global temperatures have increased by around 1.5°C, which has led to an increase in the frequency and severity of natural disasters (IPCC, 2017).

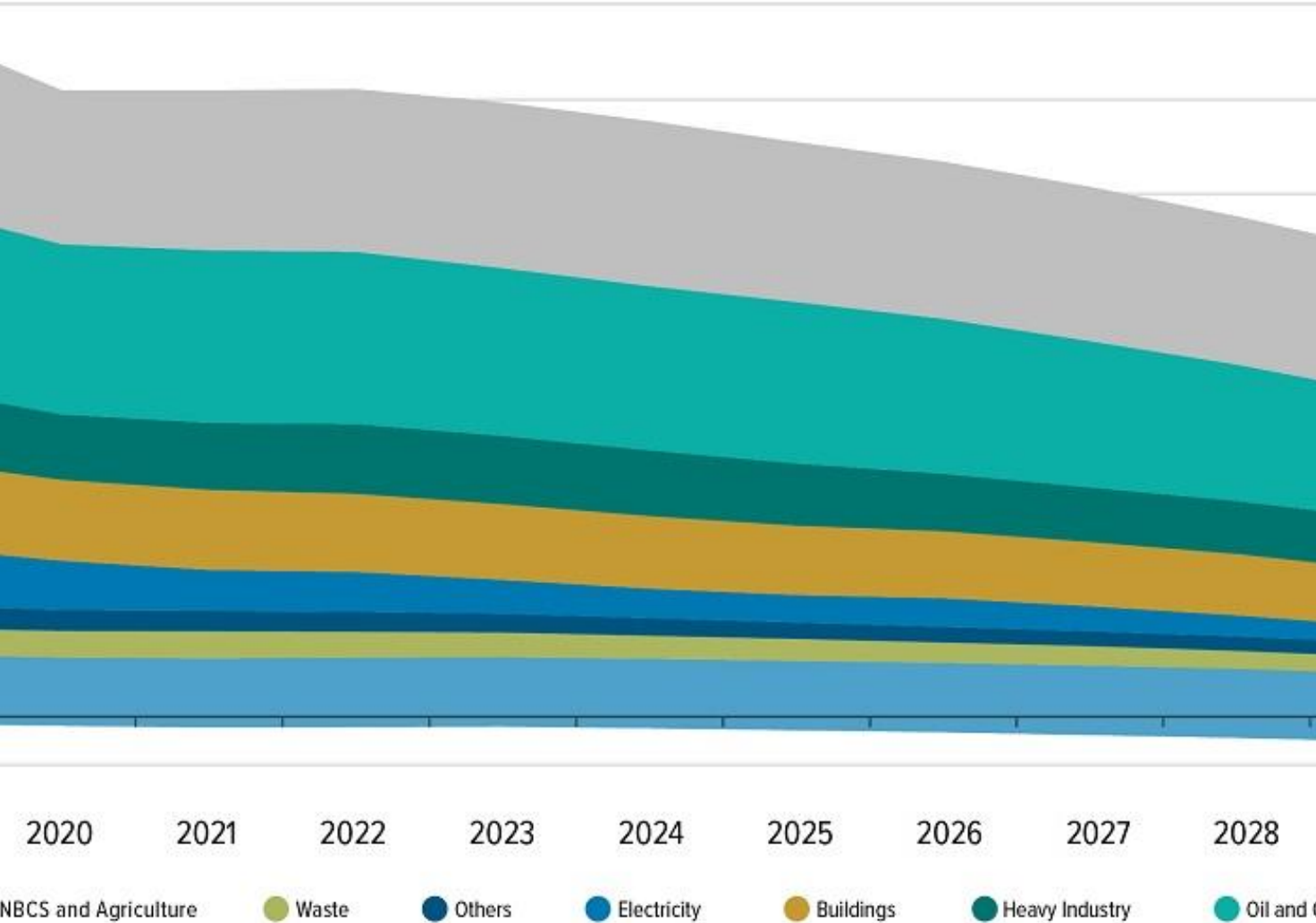




PROBLEM DESCRIPTION: Canadian Situation

The effects of climate change are significant in Canada. In 2022, the nation emitted 707.767 megatonnes of greenhouse emissions equal to carbon dioxide (Environment and Climate Change Canada, 2023). Canada saw one wildfire and two significant floods in 2020 alone. According to the Government of Canada, these disasters are classified as extreme if they satisfy certain requirements, like killing ten or more people, injuring one hundred or more, requiring national aid, or causing enough damage that the local community is unable to repair on its own

Pathway to 2030



PROBLEM STATEMENT

Canada's existing policies are expected to fall short of the nation's 2030 emission reduction goals by approximately 100 million tons of CO2 equivalent, indicating that the country is considerably behind its climate action ambitions.

PROBLEM STATEMENT:

Understanding the Impact and Consequences of the Issue



CLIMATE CHANGE HAS LED TO AN INCREASE IN THE FREQUENCY AND SEVERITY OF EXTREME WEATHER EVENTS, INCLUDING FLOODS AND WILDFIRES, IN CANADA.



THESE CATASTROPHES IMPOSE A TERRIBLE TOLL ON PEOPLE'S LIVES, RESULTING IN FATALITIES, EVICTIONS, AND SEVERE FINANCIAL LOSSES FROM WHICH COMMUNITIES OFTEN NEVER FULLY RECOVER.



IT IS NOT YET THE CASE THAT CANADIANS ARE REALLY CONCERNED FOR THEIR HEALTH AND SAFETY DUE TO A LACK OF PUBLIC PERCEPTION AND AWARENESS.

PROBLEM STATEMENT:

Analyzing the Shortcomings of Current Interventions

Regional and provincial differences in belief on the matter make it very difficult to create a country wide plan to tackle climate change

Canada's economic specialties fall within oil and gas production as we're rich in resources. Taking away or penalizing this industry too much will greatly affect our GDP.

The absence of thorough implementation and enforcement of current initiatives is preventing significant progress towards climate targets.

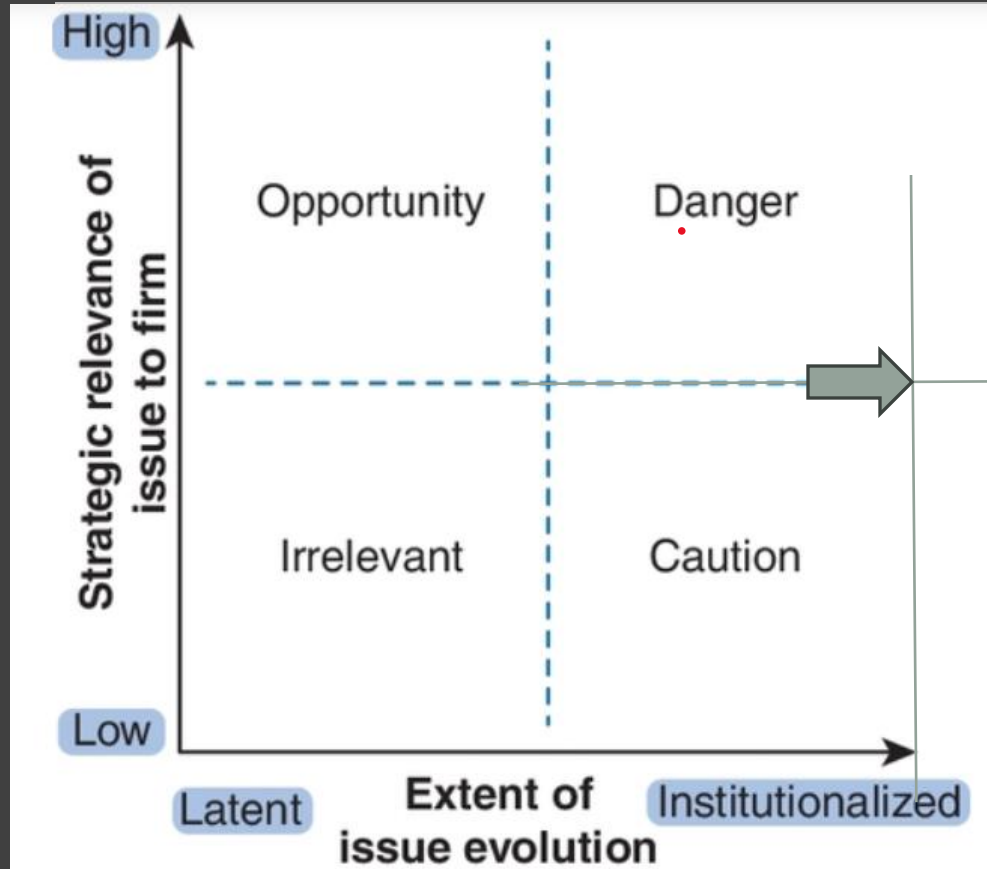
The federal, provincial, and local governments are not coordinating well enough, which leads to disjointed and uneven climate action.



IDENTIFYING THE STAKEHOLDERS

Stakeholders	Classification
Government Bodies	Organizational
Indigenous Communities	Societal
Environmental Organizations	Organizational
Financial Institutions	Economical
Other Trade Agreement Countries	Organizational
Citizens	Societal

PRIORITIZE CLIMATE ACTION



The issue of climate change would fall under the more dangerous quadrant within the table. Meaning that the extent of the issue is very institutionalized as well as high in relevance to the firm (in this case the Country of Canada). It's an issue that should be taken seriously and put high on the priority list to get a handle on

Stakeholders	Priority Based on Motivation to Act and Potential Operational Impact.
Environmental Organizations	1st: highest overall motivation to act as it is their purpose for existing and they would have a high impact as their funds are directly used for solutions to initiatives such as climate action.
Government Bodies	2nd: highest overall potential operational impact, governments have the most power to enact change.
Citizens	3rd: high motivation to act and as a collective citizens have a strong influence.
Financial Institutions	4th: most financial institutions have CSR initiatives and have the funds to have a large operational impact.
Indigenous Communities	5th: largely affected by climate change and the issues that come with it, but they have little power to enact change.
Other Trade Agreement Countries	6th: have the least interest in Canadian issues but may help as part of the trade agreement.

PRIORITIZE THE STAKEHOLDERS



PRIORITIZE THE STAKEHOLDERS

PART 2: Climate Action

HIMAYA GAMAGE, CHARLOTTE CYR, AARON FULOP

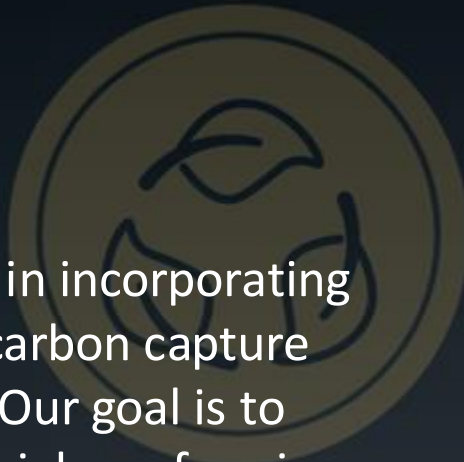
INTRODUCING GCF

Who we are:

GCF is leading the way in incorporating cutting-edge artificial carbon capture technology into cities. Our goal is to use visually beautiful, high-performing green solutions to reduce Canadian carbon emissions. A Canadian based company aimed to helping Canadians and their goals.

What we do:

We design and manufacture artificial carbon capture trees that are able to absorb 1000x faster the amount of carbon dioxide compared to natural methods.





SOLUTION DESCRIPTION

GCF offers cutting-edge Artificial Carbon Capture Trees (ACCT) technology that replicates the organic carbon sequestration process carried out by actual trees, thereby mitigating the effects of climate change. The most important differentiating factor is their bigger and more effective capacity to absorb carbon dioxide (CO₂) from the atmosphere than natural trees. With their scalability, these artificial trees offer a promising way to lower greenhouse gas emissions and slow global warming, giving you reason to be optimistic about their potential impact. We'll be working with countless companies nationwide to provide our CCT to reduce the amount of CO₂ produced in our air. In contrast to natural trees, which require years to reach maturity and are vulnerable to pests, diseases, and bad weather, ACCT units sequester carbon instantly upon installment, offering consistent and dependable carbon sequestration.

GCP provides Artificial Carbon Capture Trees that can be far more efficient in absorbing CO₂, providing a quick and dependable reduction in carbon emissions to fight global warming.

MISSION STATEMENT

Reduce carbon emissions in Canada using artificial carbon capture trees.



ELEMENTS OF IMPACT BUSINESS MODEL: Target Market

TAM (assuming all potential customers are reached)

- 38 million people globally
- Urban, suburban, rural areas across all provinces and territories
- Residential, commercial, industrial, transportation and agricultural sectors

SAM

- Canadian urban regions with high pollution levels, provinces with progressive climate policies
- Canadian companies trying to reach their sustainability goals
- Canadian municipalities that want to put green infrastructure into place

SOM

- Environmentally aware consumers and companies
- Cities such as Guelph

ELEMENTS OF IMPACT BUSINESS MODEL: Demographic and lifestyle characteristics

With a population of about 150,000, Guelph is steadily expanding. Due in large part to the University of Guelph student population, the population is particularly young, having a large percentage of individuals aged 18 to 34. Additionally, the city has a sizable working-age population (35–54) and a growing elderly population (55+) (City of Guelph, 2021). The lifestyle of Guelph places a strong emphasis on environmental sustainability, community involvement, and quality of life. The city has many parks, recreational areas, and a thriving cultural scene. The importance of sustainability to the locals is demonstrated via conservation and green projects. The vibrant sense of community is promoted by festivals, farmers' markets, and local events. Guelph is a perfect market for artificial carbon capture trees because of its demographic and lifestyle characteristics, which complement the city's sustainability objectives and core values.

ELEMENTS OF IMPACT BUSINESS MODEL:

ADOPTION

- Enormous tree planting and manufacturing initiatives can lead to job openings in maintenance, forestry, and landscaping.
- Artificial trees offer a regular and dependable way to trap carbon dioxide, unlike natural trees, which are susceptible to diseases, pests, and climate change.
- Artificial trees are not just reliable but also impressively efficient. They can capture CO₂ more quickly and in greater amounts than regular trees, a testament to their performance in carbon capture. ACCT can be designed to take up less space, making them ideal for urban environments and areas with limited land availability

REJECTION

- Given that ACCT technology is still in its early stages, there is a pressing need for further long-term studies and data to fully comprehend its efficacy, drawbacks, and long-term effects on the environment.
- Some communities may not find artificial trees visually pleasing, particularly in places with strong cultural or aesthetic linkages to natural environments.



ELEMENTS OF IMPACT BUSINESS MODEL: Competition and Competitive Advantage

Factors of Success in the Industry

- Low cost
 - Produced in Canada to reduce shipping emissions and cost
- Effective product
 - Technology of the future for the future of our environment
- Canadian company built with Canada in mind
 - Our values align with those of Canadians
- Create jobs for Canadians
 - Based in Canada and run by Canadians

ELEMENTS OF IMPACT BUSINESS MODEL: Competition and Competitive Advantage

Identify your competition

- The Status Quo
 - Doing nothing can be a company's undoing in comparison to its competition which has a social responsibility plan
- Government



ELEMENTS OF IMPACT BUSINESS MODEL: Competition and Competitive Advantage

STRENGTHS

- Technology is already established
 - The technology has already been created in the USA
- Large market for Carbon Capture Trees in Canada
 - No other company has a product like ours
- Lower cost
 - No shipping cost
- Reduced emissions
 - Can absorb carbon 1000 times faster than a natural tree
- No competitors
 - There are no other Canadian carbon capture tree companies

WEAKNESSES

- First to do it in Canada
 - There may be obstacles to producing CCT's, and we will be the first to encounter those
- Product is not well known
 - There is little public information on this technology, so we will have to get the word out
- Start-up cost
 - There is no facility for building CCT's in Canada yet
- No established clients

ELEMENTS OF IMPACT BUSINESS MODEL: Competition and Competitive Advantage

OUR OFFER

A Canadian-based company committed to easing climate change through the reduction of carbon emissions and greenhouse gases by producing Artificial Carbon Capture Trees.

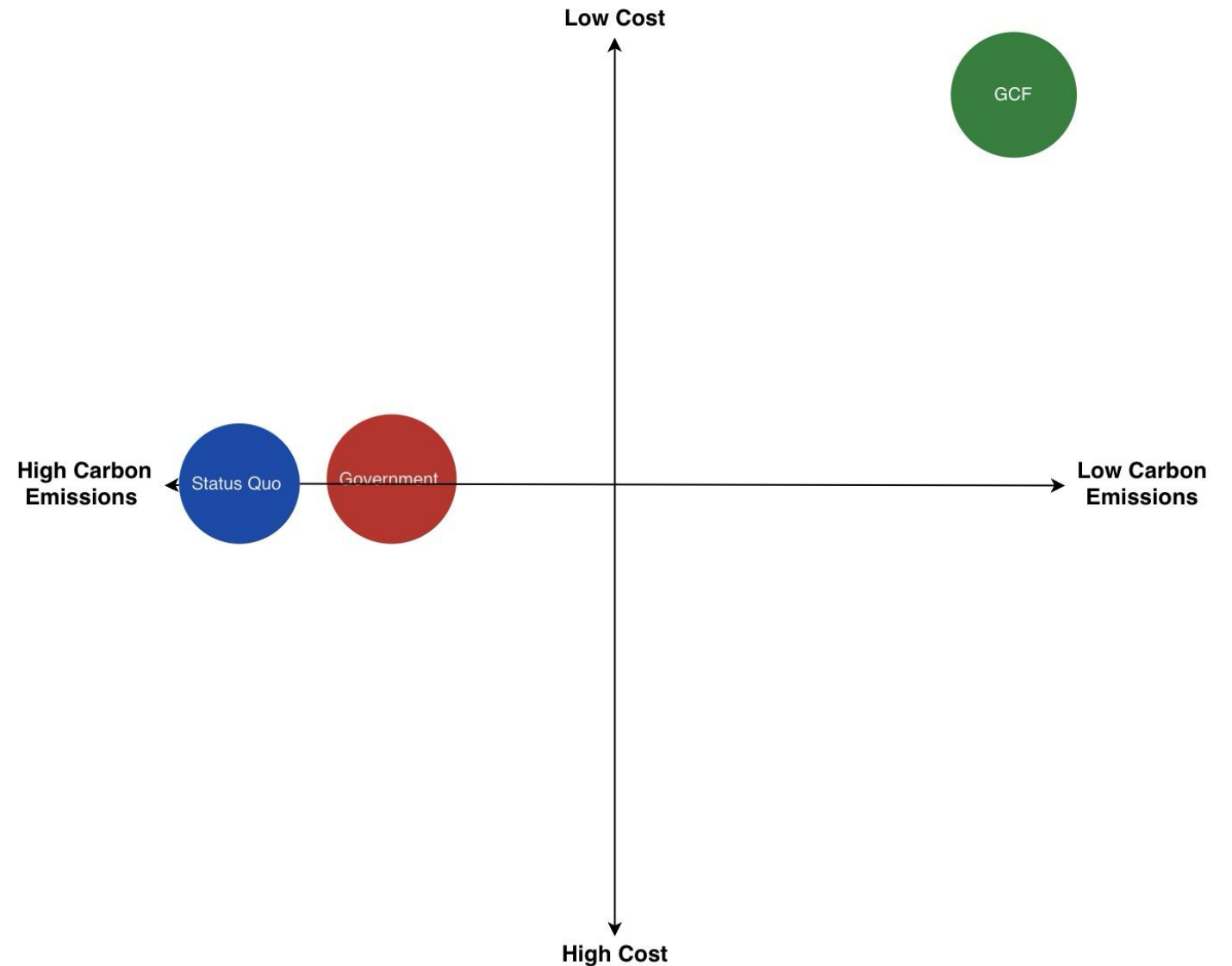
COMPETITIONS OFFER

The only competition is the status quo or the Canadian government who have various incentives and programs to guide business' to reduce their own carbon footprint. Which include rebates, tax credit for those who wish to participate in a greener solution.



ELEMENTS OF IMPACT BUSINESS MODEL: Competition and Competitive Advantage

Perceptual map of the
competition



ELEMENTS OF IMPACT BUSINESS MODEL: Competition and Competitive Advantage

Why choose GCF?

Our company is the only Canadian company committed to helping Canadians through Carbon Capture Trees. Our product is 1000 times more effective than natural trees at absorbing Carbon from the atmosphere. We offer a low-cost high-impact solution that could greatly improve the lives of Canadians by slowing climate change and moving toward a zero-emission future. There is a moral and social obligation for entities to act as awareness of climate change and its effects grows. We care about your dedication to cutting carbon emissions and strive to help you towards that goal.



ELEMENTS OF IMPACT BUSINESS MODEL: Value Proposition

For (target user) - Urban municipalities, large businesses and government bodies

Who wants/needs (target application) - for corporations that emit large amounts of CO₂ and governments who are concerned about the amount of CO₂ being emitted.

Our product (statement of primary differentiation) - The GCF Artificial Carbon Capture Trees (ACCTs) is a green technology solution. Our artificial carbon capture trees are redefining environmental stewardship with their unmatched efficiency in reducing CO₂ emissions. We offer scalable solutions that redefine sustainability using cutting-edge technology, guaranteeing future generations will live in a more environmentally friendly world. The Artificial Carbon Capture Tree offers better performance and aesthetic benefits without looking large or less integrated into urban landscapes than current CO₂ removal systems. Instead, it blends in flawlessly.





ELEMENTS OF IMPACT BUSINESS MODEL:

Value Proposition

How we differ from the competition

- Our trees trap CO₂ at far higher rates than current methods, assuring maximum environmental effect. This is achieved by utilizing cutting-edge materials and creative design.
- Our trees may be effortlessly incorporated into a variety of landscapes, from city parks to industrial areas, without sacrificing performance, as they are designed for both urban and rural environments.
- Our carbon capture trees require very little external energy inputs to function, in contrast to many of our competitors. In the long run, this makes them a more economical and sustainable solution.
- Our trees are made to last a long time with little upkeep needed, which lowers operating expenses and downtime. They are constructed with sturdy, weather-resistant materials.

A photograph of a single, full-canopied tree with green leaves standing in a grassy field. In the background, a city skyline is visible under a clear blue sky. The tree is positioned on the left side of the frame, and the city skyline spans the horizon.

ELEMENTS OF IMPACT BUSINESS MODEL: Product Positioning

Artificial Carbon Capture Trees

Our technology will allow more CO₂ to be captured than natural trees. This will be suitable for the various environments in Canada, ranging from urban to industrial sites. A single tree can capture one tonne of carbon dioxide daily, allowing Canada to reduce its overall emissions. Since a single tree can capture such a large amount, integrating them into large cities will be accessible as a small number is required.

ACCT technology offers immediate, dependable, and scalable benefits to a wide range of stakeholders dedicated to environmental sustainability and climate action. Its scalability means that its impact can grow alongside the increasing need for carbon sequestration, providing a hopeful outlook for the future.

ELEMENTS OF IMPACT BUSINESS MODEL:

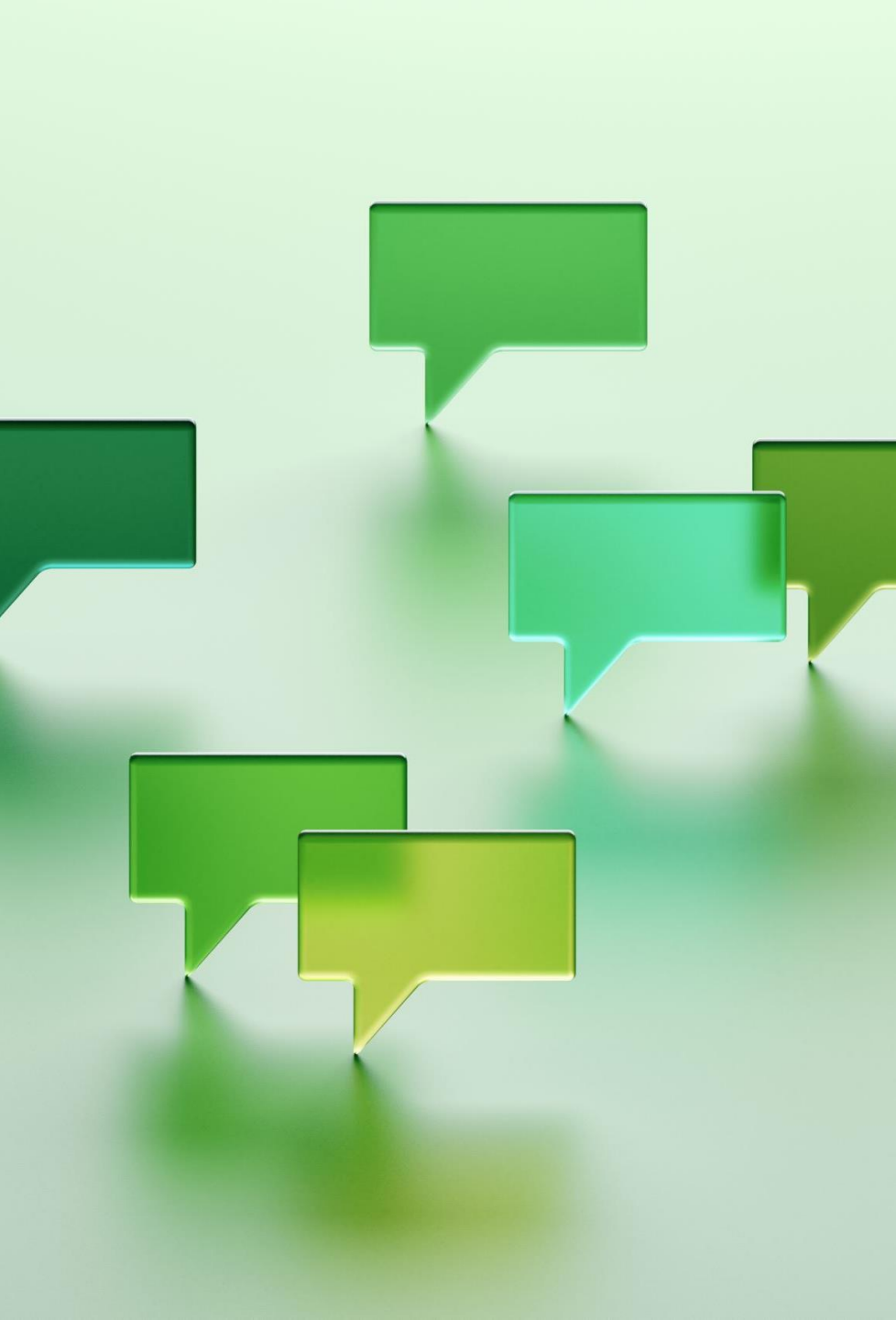
5 P's - Price and Revenue Stream

The installation size, personalization choices, and continuing maintenance will all play a role in determining the pricing plan. Depending on these factors, the price per unit could vary from \$5,000 to \$50,000.

Stream of Revenue:

- The money made from selling artificial carbon capture trees to companies, organizations, and towns directly.
- Fees for subscription/maintenance services: Constant income from contracts for upkeep and subscription services for regular updates and performance tracking.
- Joint ventures for co-branding and sponsorships with businesses, environmental groups, and universities.





ELEMENTS OF IMPACT BUSINESS MODEL: 5 P's - Promotion

- Use sites like Facebook, Instagram, Twitter, and LinkedIn to disseminate exciting content about the advantages of carbon capture trees to a broad audience.
- Reach out to prominent news sources with press releases and negotiate features in blogs and publications devoted to sustainability and the environment.
- Hold seminars, websites discussing the advantages and effects of carbon capture trees.
- Continual updates on upcoming initiatives, their effects on the environment, and ways to become involved.
- Collaborate with Canadian universities (such as University of Guelph) to study and assess the performance of carbon capture trees while offering student research, internships or thesis opportunities.



ELEMENTS OF IMPACT BUSINESS MODEL: 5 P's - Place

Channels of Distribution:

- Via the company's online store and specialized sales staff.
- To reach a wider audience, work with landscapers, businesses and municipalities.
- Collaborate with other companies in urban planning and green technology

These distribution channels are designed to be inclusive, ensuring wide accessibility and practicality for all customer categories, from big businesses to private corporations.

A large, light blue puzzle piece is positioned on the left side of the slide, set against a solid yellow background. The puzzle piece is oriented diagonally, with its top-left corner pointing towards the upper left. It has several interlocking tabs and blanks, suggesting it is part of a larger assembly. The lighting creates a soft shadow to the right of the piece, giving it a three-dimensional appearance.

ELEMENTS OF IMPACT BUSINESS MODEL:

5 P's - Partner

Municipalities and Urban Planners

- Work together to include trees into city infrastructure projects by collaborating with municipal administrations and urban planners.

Environmental NGOs

- Collaborate with them to spread the word about the technology and take advantage of their networks.

Technologies Providers

- For technological help and integration, collaborate with businesses that specialize in smart infrastructure and green technologies.

Corporate Sponsors

- To increase awareness and financial prospects, approach companies who are interested in sustainability to sponsor installations or co-brand projects.

Problem Statement Canada's existing policies are expected to fall short of the nation's 2030 emission reduction goals by approximately 100 million tons of CO2 equivalent, indicating that the country is considerably behind its climate action ambitions.				
Mission Statement Reduce carbon emissions in Canada using artificial carbon capture trees				
Key Partners	Key Activities Not required for assignment	Value Proposition Advanced Efficiency Scalability and Flexibility Energy Independance	Stakeholder Relationships Not required for assignment	Stakeholder Segments Include TAM, SAM, SOM TAM Municipalities Large businesses Educational Institutes
Government Agencies Research Institutions Energy Companies Financial Institutions Tech Providers	Key Resources Not required for assignment	Durability and Low Maintenance	Channels (Include positioning, promotion, place) Green technology events- Place Social Media - Promotion Interview with News Outlets- Promotion Collaborate with Canadian Universities to conduct research - Positioning	SAM Canadian municipalities, businesses, environmental organizations SOM City of Guelph University of Guelph Local businesses
Cost Structure Not required for assignment			Revenue Streams Price per unit can range from \$5000-\$50,000 Revenue comes from upfront sale as well subscription fees for recurring maintenance	

intended Impact
(Assume your venture has been successfully operating for 1 year and everything is according to plan, what change would you like to see in the statistics of your SOM in Guelph)
Our planned impact, following a year of successful operation of the Artificial Carbon Capture Trees (ACCT) enterprise, is a 5% decrease in CO2 emissions in the locations where ACCT units are installed in Guelph. This would be accomplished by installing 10 ACCT devices, which would greatly increase the city's capacity to sequester carbon dioxide. Each ACCT unit would capture one ton of CO2 yearly. Through educational initiatives and neighbourhood gatherings, we hope to increase public knowledge and engagement by 50%. From an economic standpoint, the project would create jobs, earn cash from carbon credits, and save \$500,000 in expenses. Our goal would be further supported by forming at least ten strategic alliances with nearby companies, governmental organizations, and non-governmental organizations. At least two published studies on the impact and efficacy of ACCT technology would be the product of partnerships with the University of Guelph, and there would be a thirty percent increase in community involvement and satisfaction as determined by surveys.

IMPACT BUSINESS MODEL CANVAS

Your Venture: GCF
Your Legal Structure
Hypothesis: For profit
social venture

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